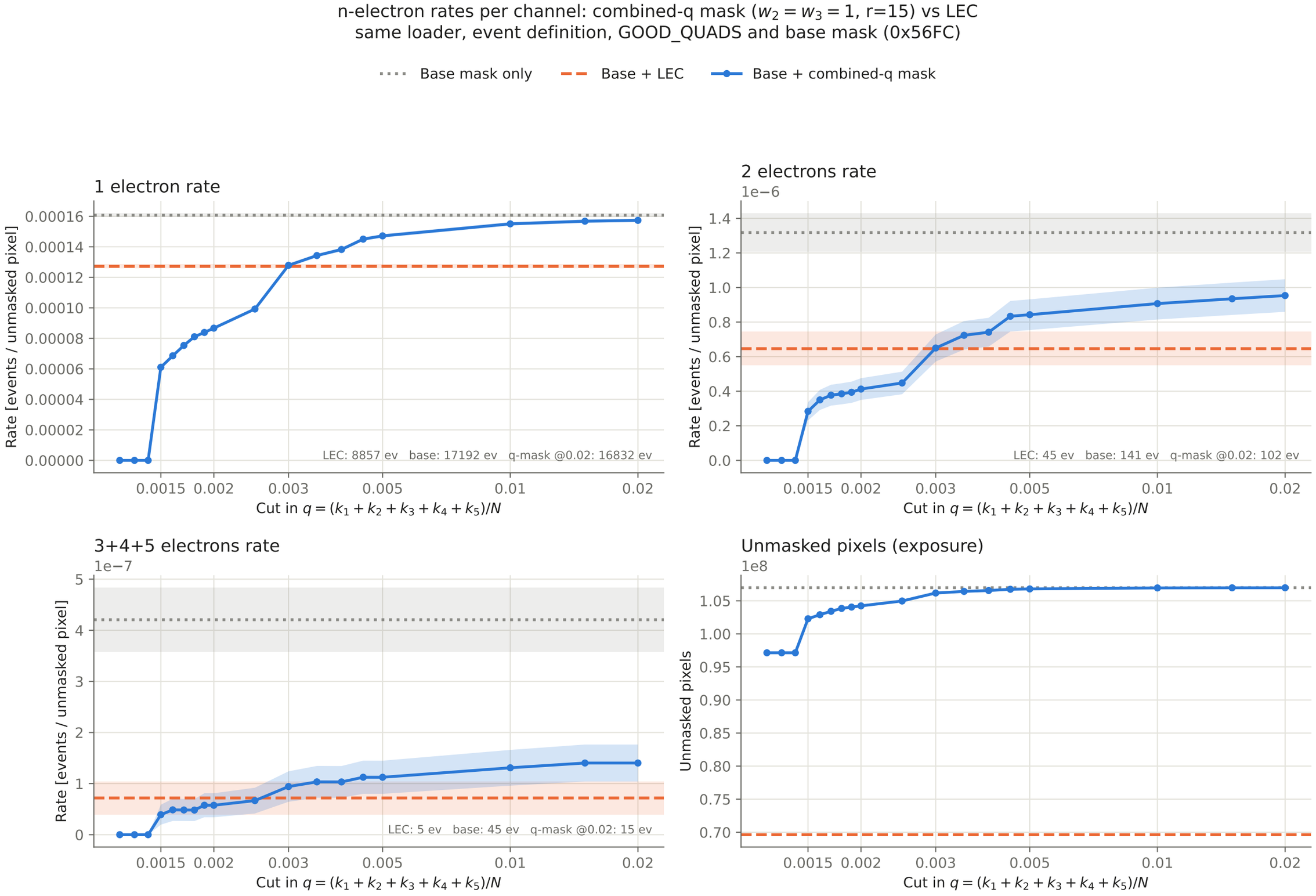
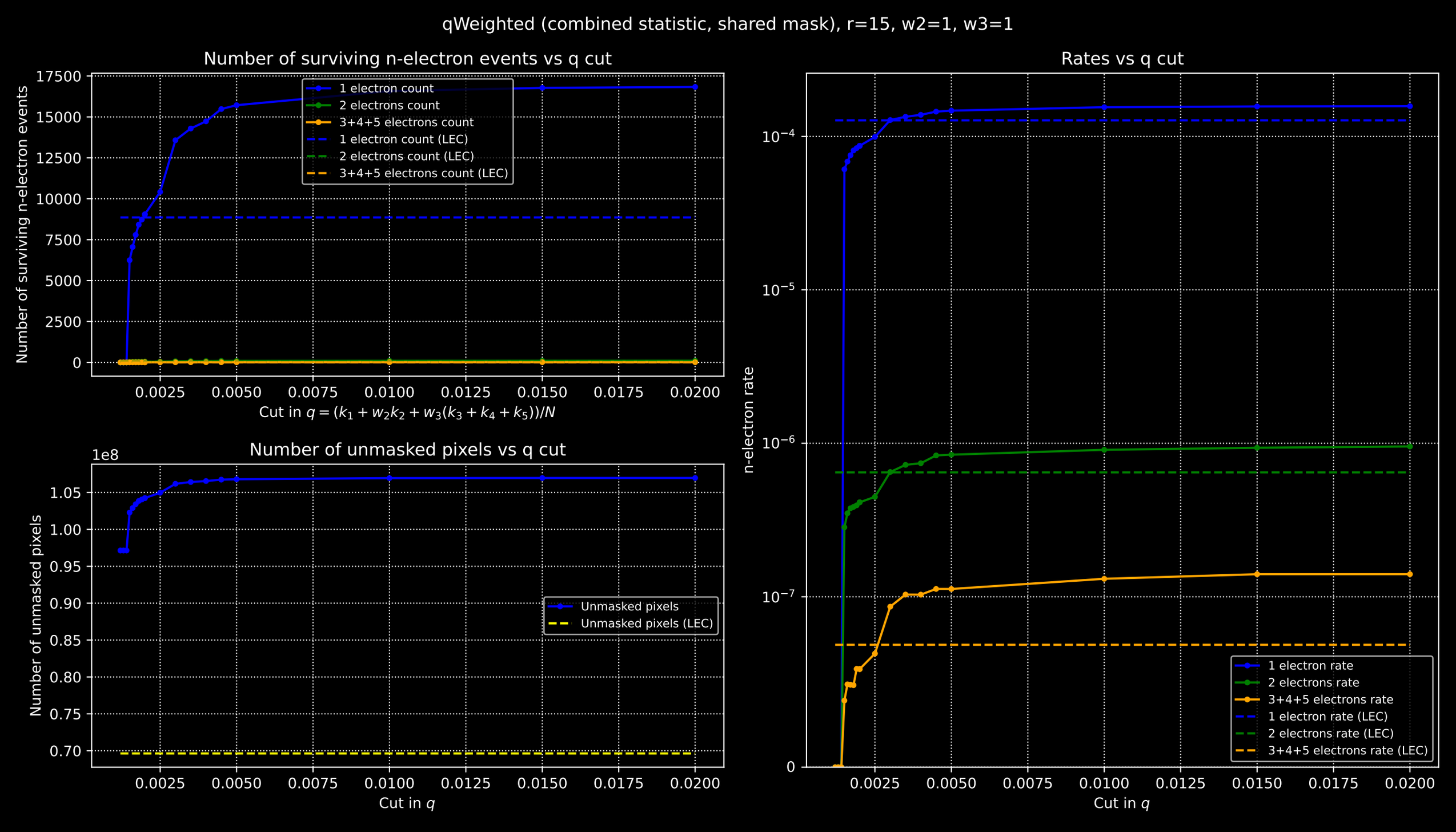
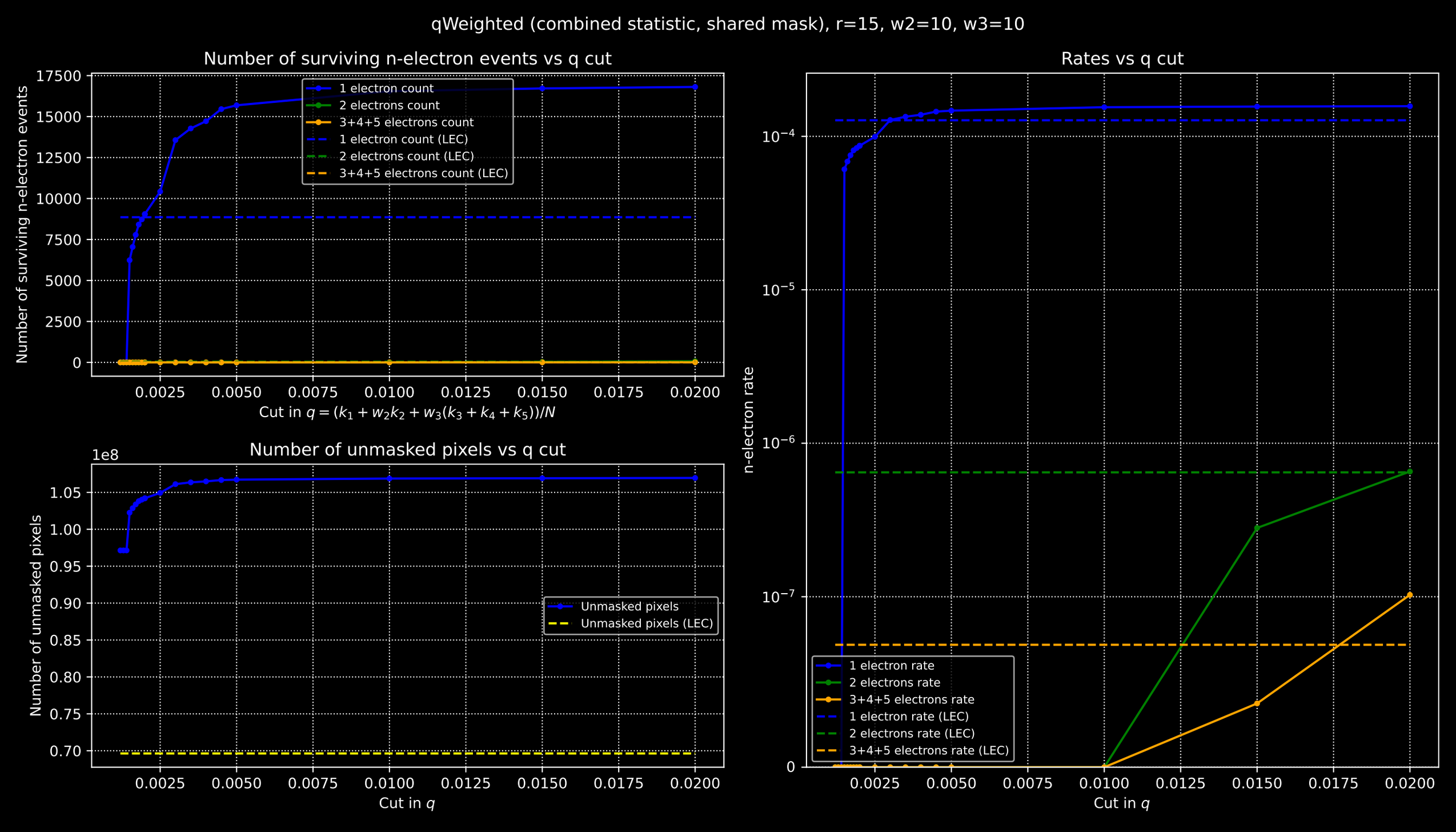
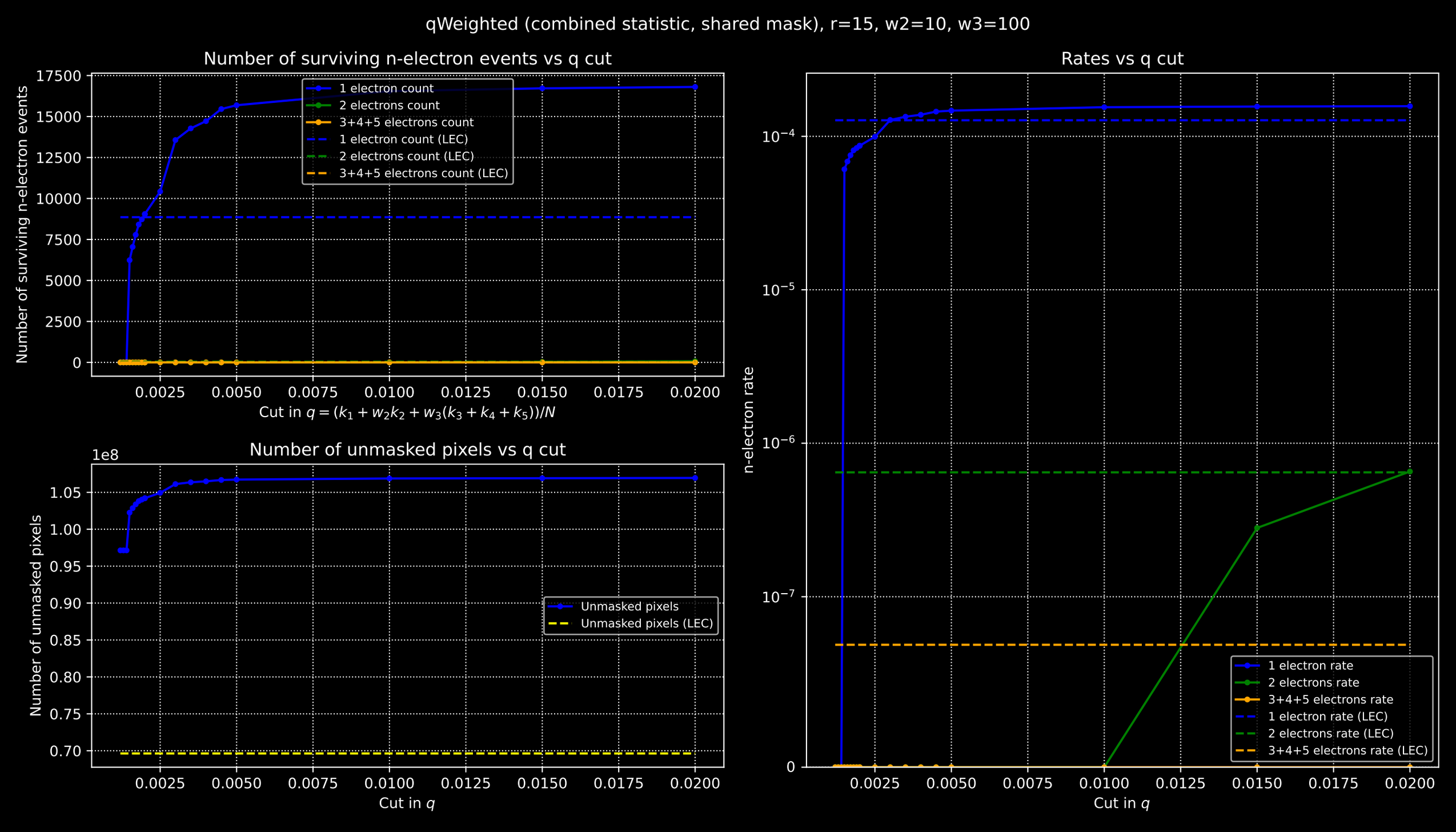
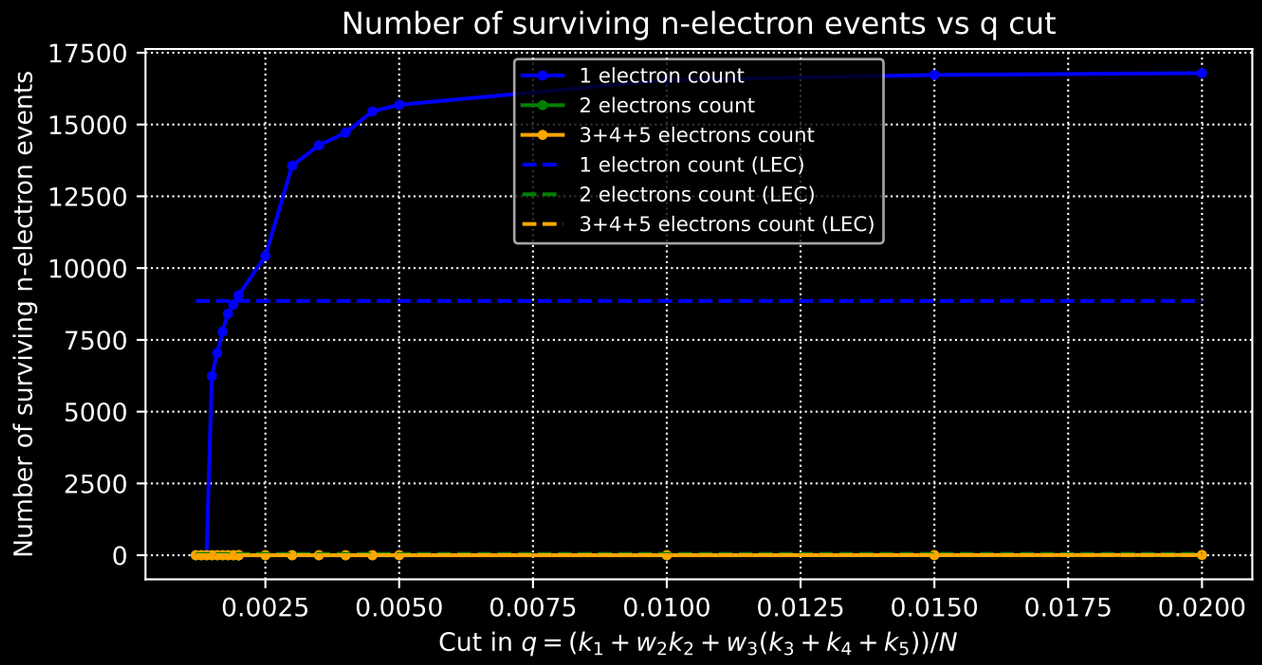

n-electron rates per channel: combined-q mask ($w_2 = w_3 = 1$, r=15) vs LEC
same loader, event definition, GOOD_QUADS and base mask (0x56FC)
Base mask only
Base + LEC
Base + combined-q mask
1 electron rate
Rate [events / unmasked pixel]
0.00016
0.00014
0.00012
0.00010
0.00008
0.00006
0.00004
0.00002
0.00000
LEC: 8857 ev base: 17192 ev q-mask @0.02: 16832 ev
0.0015 0.002 0.003 0.005 0.01 0.02
Cut in $q = (k_1 + k_2 + k_3 + k_4 + k_5)/N$
2 electrons rate
1e−6
Rate [events / unmasked pixel]
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
LEC: 45 ev base: 141 ev q-mask @0.02: 102 ev
0.0015 0.002 0.003 0.005 0.01 0.02
Cut in $q = (k_1 + k_2 + k_3 + k_4 + k_5)/N$
3+4+5 electrons rate
1e−7
Rate [events / unmasked pixel]
5
4
3
2
1
0
LEC: 5 ev base: 45 ev q-mask @0.02: 15 ev
0.0015 0.002 0.003 0.005 0.01 0.02
Cut in $q = (k_1 + k_2 + k_3 + k_4 + k_5)/N$
Unmasked pixels (exposure)
1e8
Unmasked pixels
1.05
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.0015 0.002 0.003 0.005 0.01 0.02
Cut in $q = (k_1 + k_2 + k_3 + k_4 + k_5)/N$

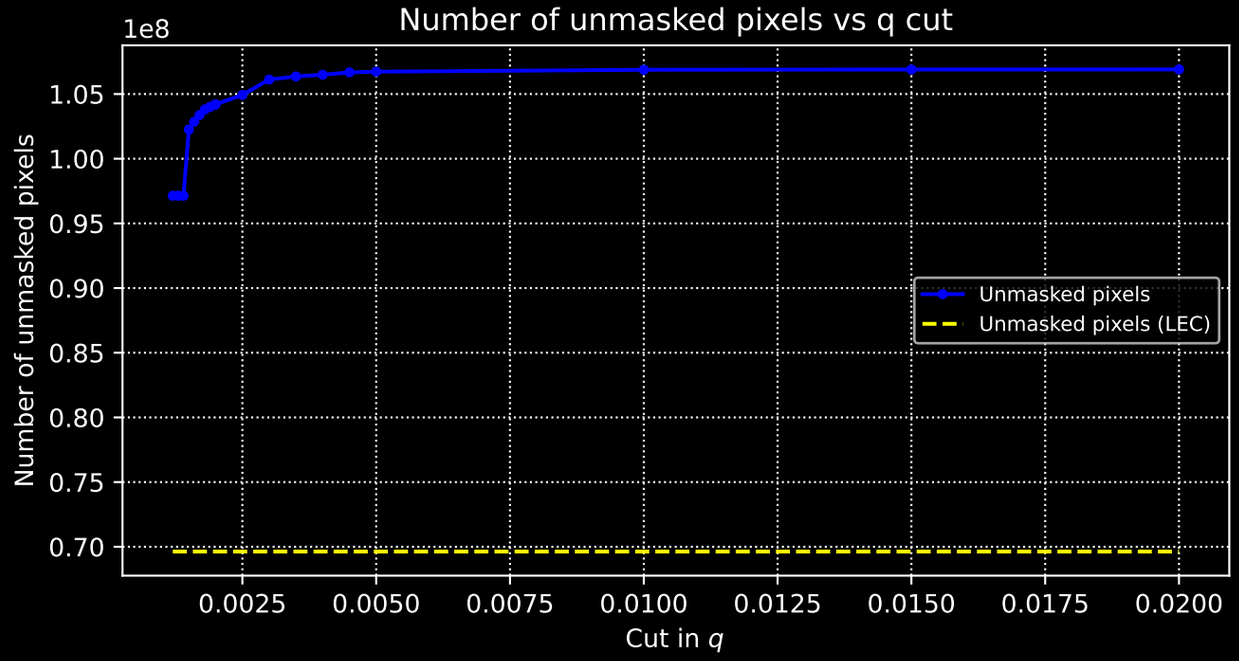

qWeighted (combined statistic, shared mask), r=15, w2=1, w3=1
Number of surviving n-electron events vs q cut
Number of surviving n-electron events
Cut in $q = (k_1 + w_2 k_2 + w_3(k_3 + k_4 + k_5))/N$
1 electron count
2 electrons count
3+4+5 electrons count
1 electron count (LEC)
2 electrons count (LEC)
3+4+5 electrons count (LEC)
Number of unmasked pixels vs q cut
1e8
Number of unmasked pixels
Cut in q
Unmasked pixels
Unmasked pixels (LEC)
Rates vs q cut
n-electron rate
Cut in q
1 electron rate
2 electrons rate
3+4+5 electrons rate
1 electron rate (LEC)
2 electrons rate (LEC)
3+4+5 electrons rate (LEC)

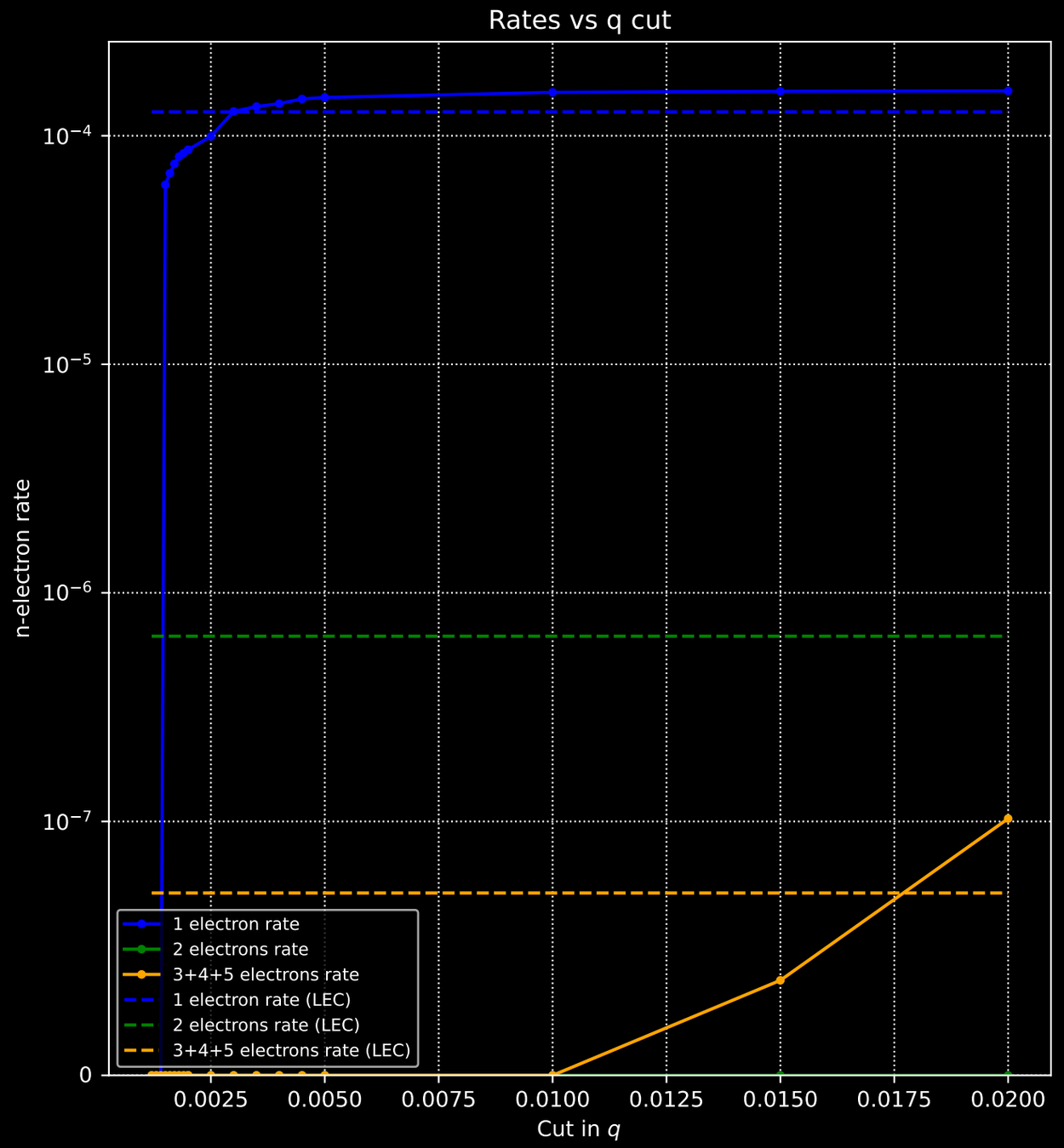

qWeighted (combined statistic, shared mask), r=15, w2=10, w3=10
Number of surviving n-electron events vs q cut
Number of surviving n-electron events
Cut in $q = (k_1 + w_2 k_2 + w_3(k_3 + k_4 + k_5))/N$
1 electron count
2 electrons count
3+4+5 electrons count
1 electron count (LEC)
2 electrons count (LEC)
3+4+5 electrons count (LEC)
Number of unmasked pixels vs q cut
1e8
Number of unmasked pixels
Cut in q
Unmasked pixels
Unmasked pixels (LEC)
Rates vs q cut
n-electron rate
Cut in q
1 electron rate
2 electrons rate
3+4+5 electrons rate
1 electron rate (LEC)
2 electrons rate (LEC)
3+4+5 electrons rate (LEC)


qWeighted (combined statistic, shared mask), r=15, w2=10, w3=100
Number of surviving n-electron events vs q cut
Number of surviving n-electron events
Cut in $q = (k_1 + w_2 k_2 + w_3(k_3 + k_4 + k_5))/N$
1 electron count
2 electrons count
3+4+5 electrons count
1 electron count (LEC)
2 electrons count (LEC)
3+4+5 electrons count (LEC)
Number of unmasked pixels vs q cut
1e8
Number of unmasked pixels
Cut in q
Unmasked pixels
Unmasked pixels (LEC)
Rates vs q cut
n-electron rate
Cut in q
1 electron rate
2 electrons rate
3+4+5 electrons rate
1 electron rate (LEC)
2 electrons rate (LEC)
3+4+5 electrons rate (LEC)

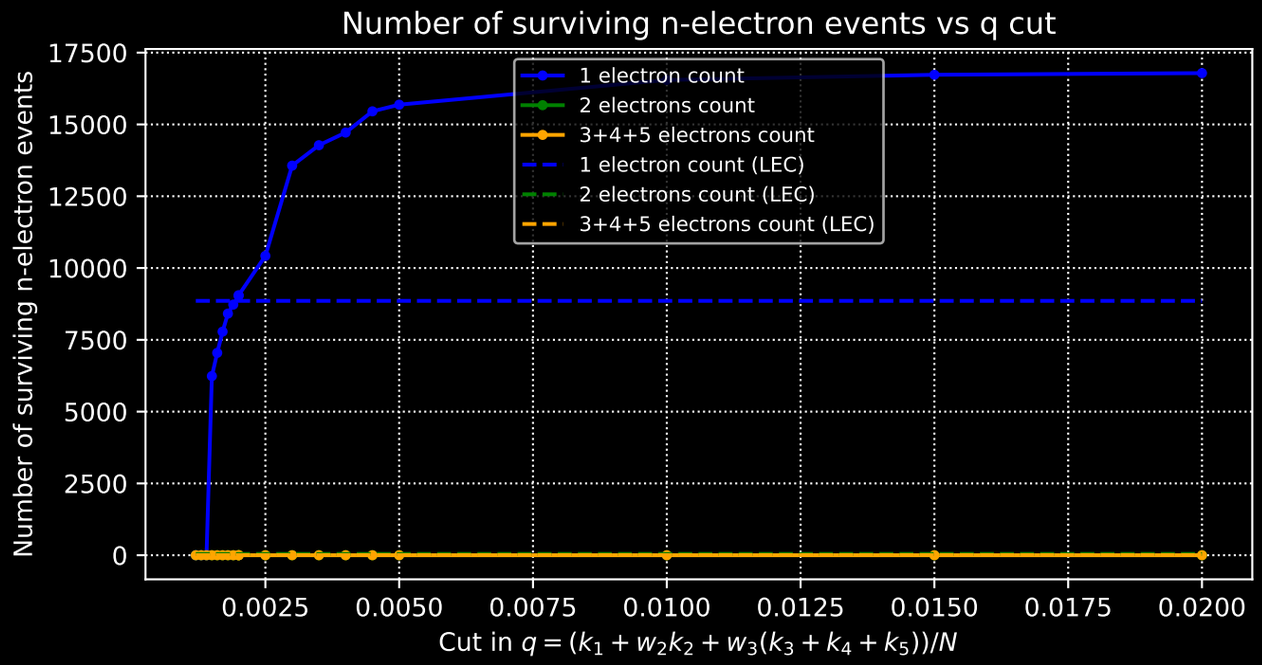
qWeighted (combined statistic, shared mask), r=15, w2=100, w3=10

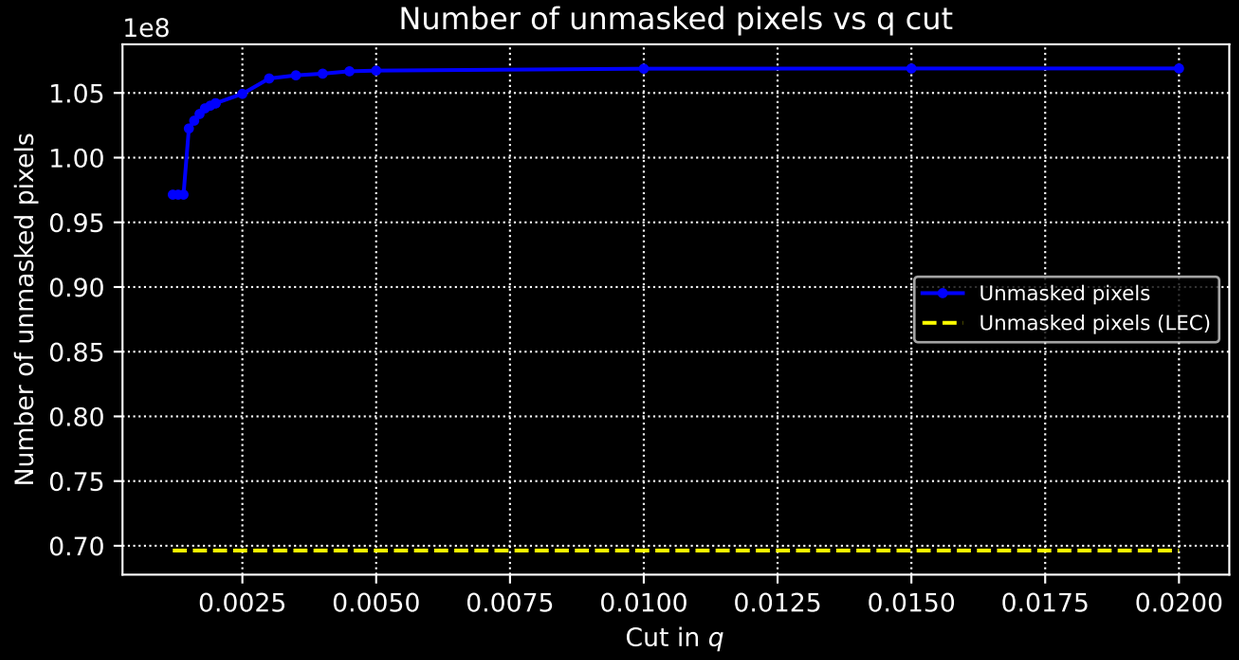


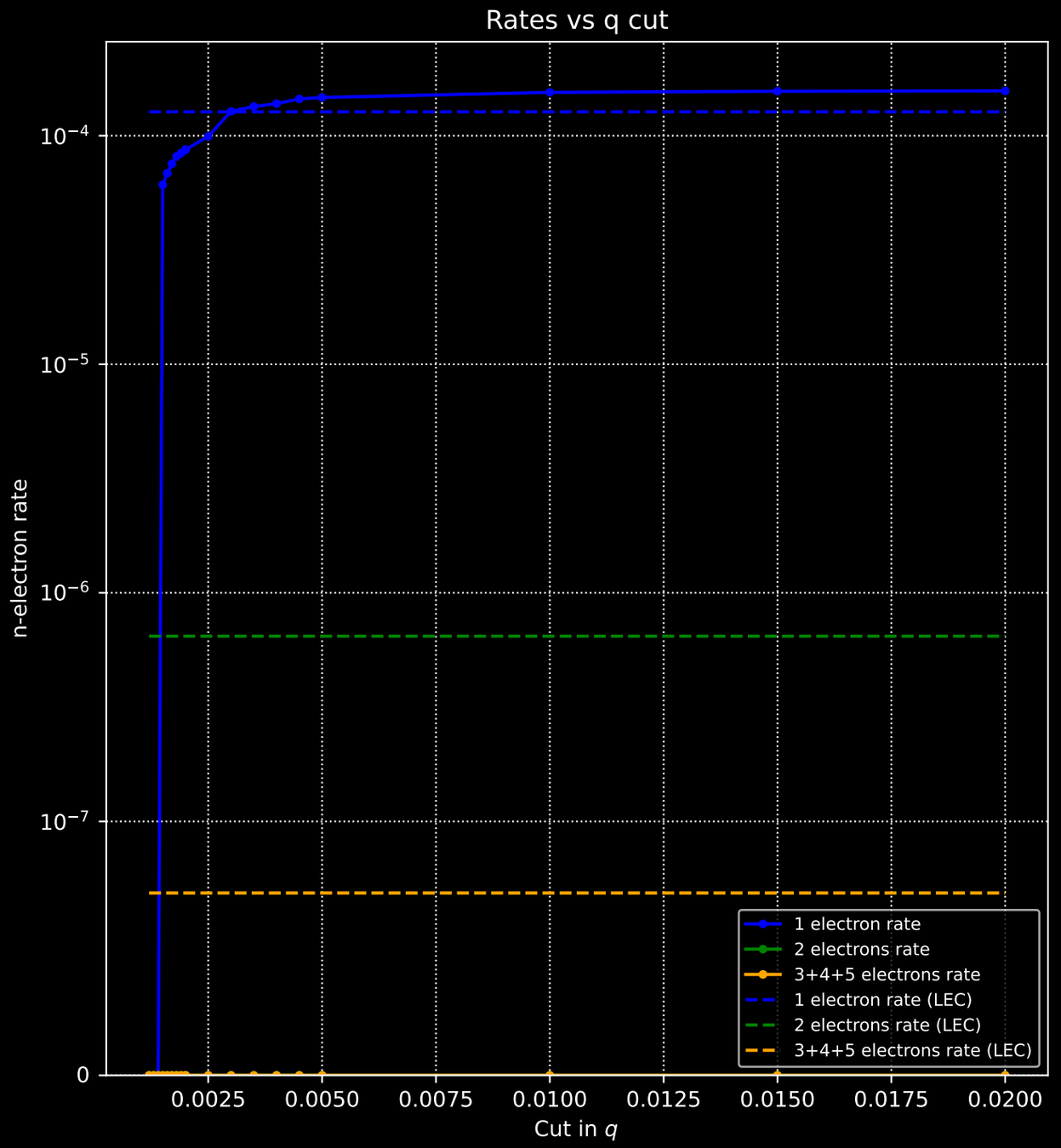




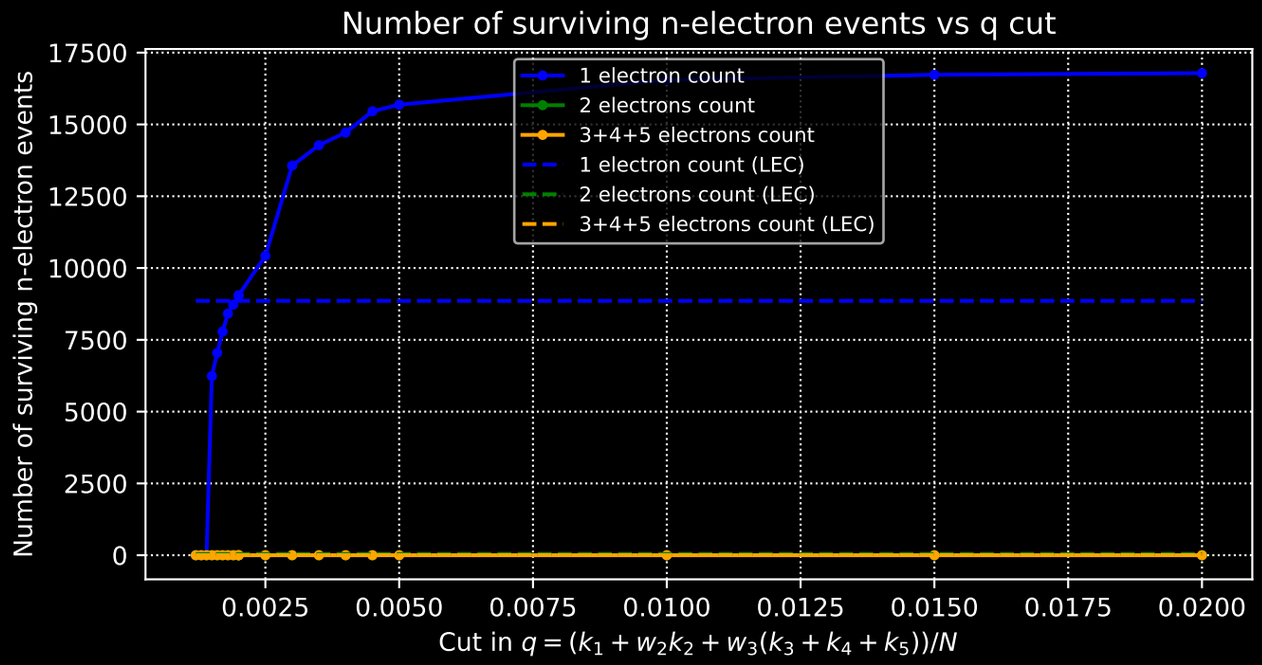
qWeighted (combined statistic, shared mask), r=15, w2=100, w3=1000

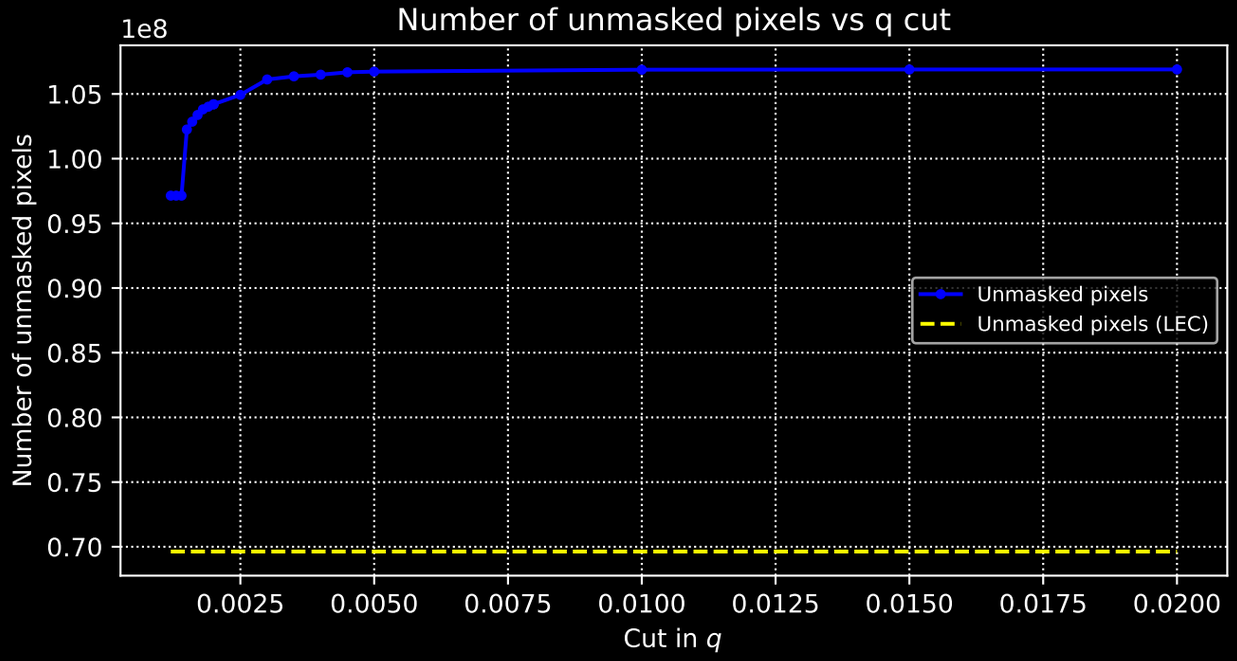


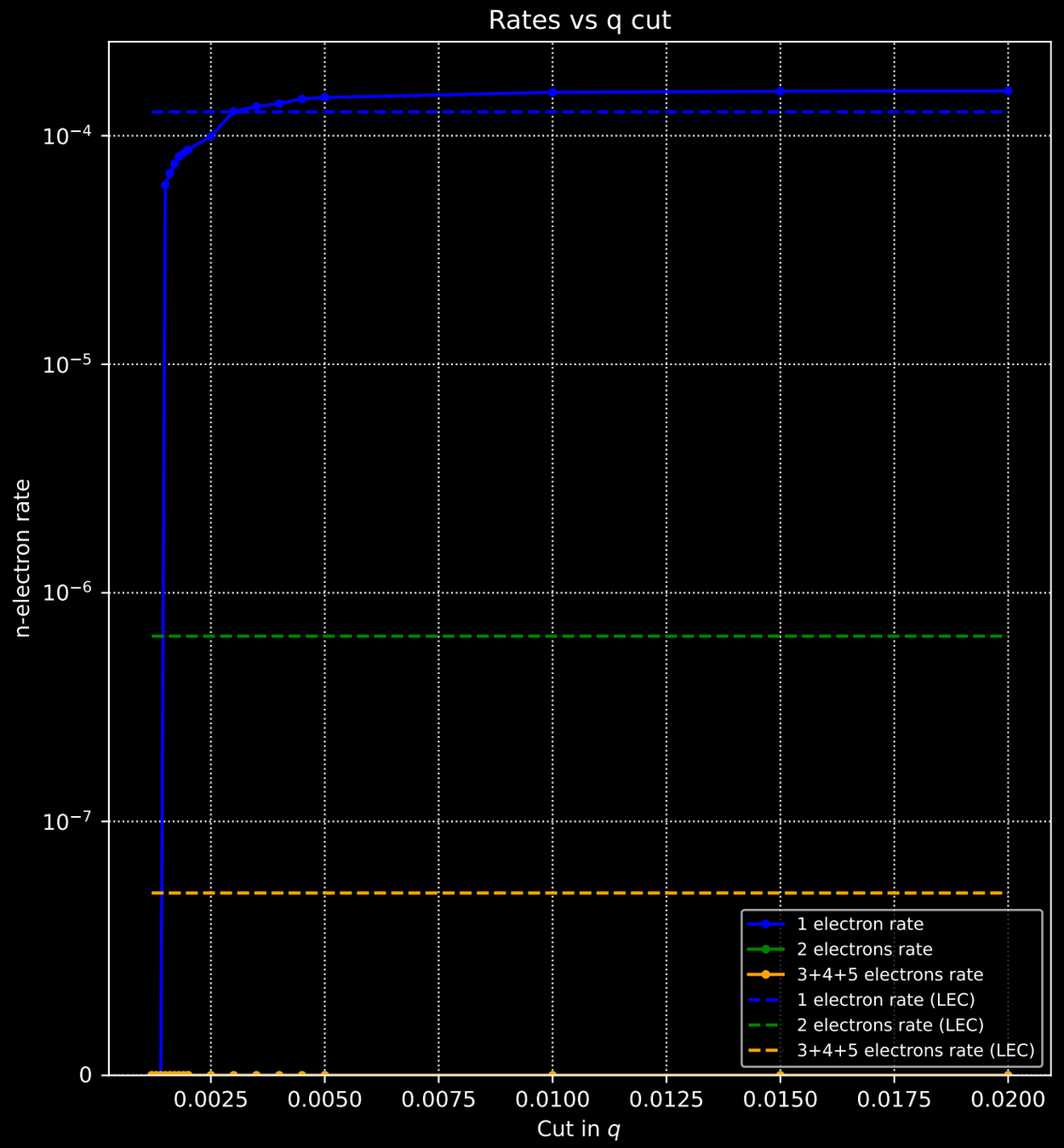




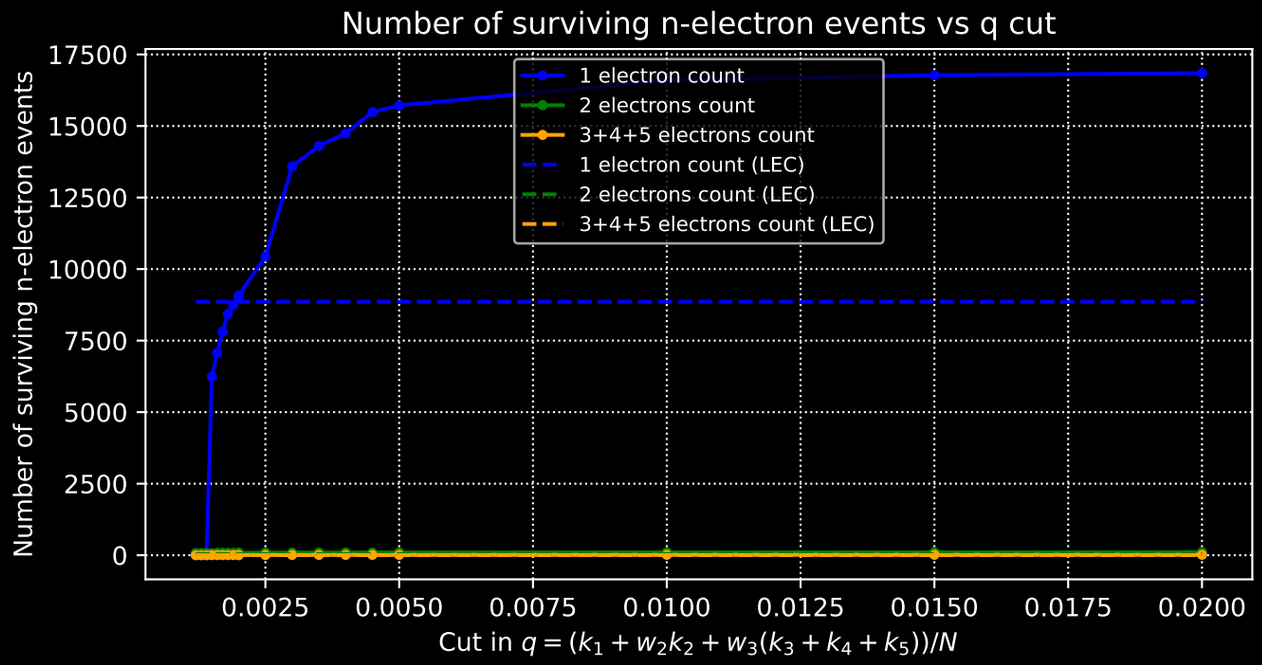
qWeighted (combined statistic, shared mask), r=15, w2=1000, w3=100

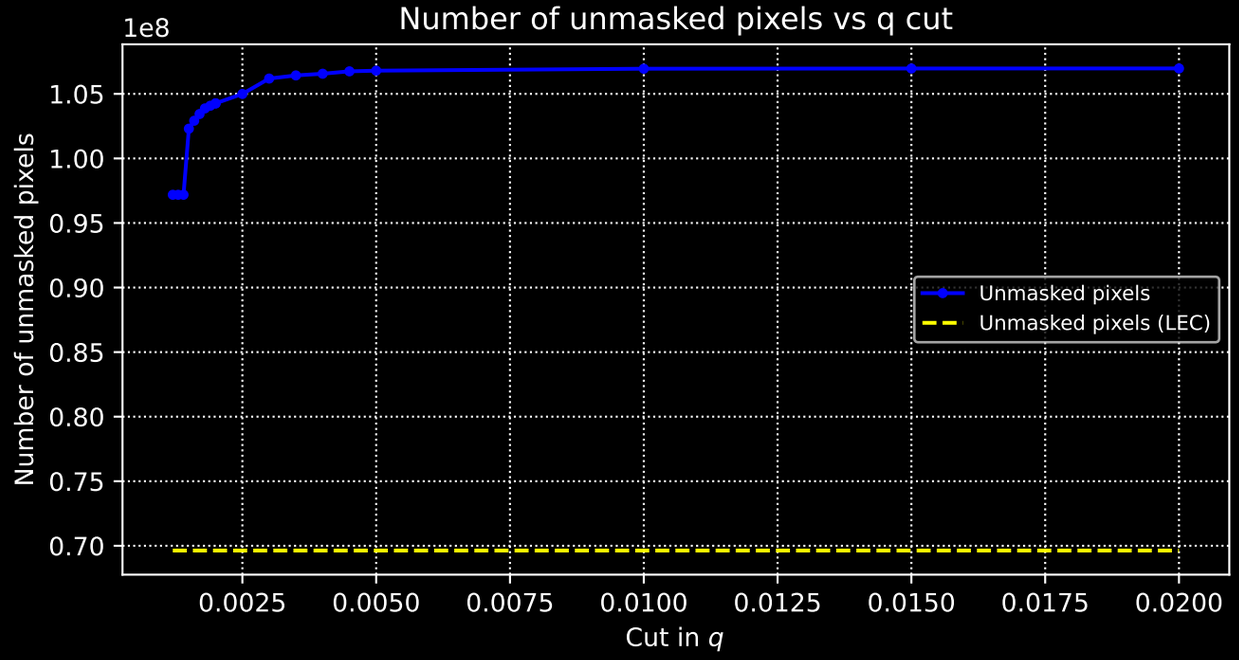


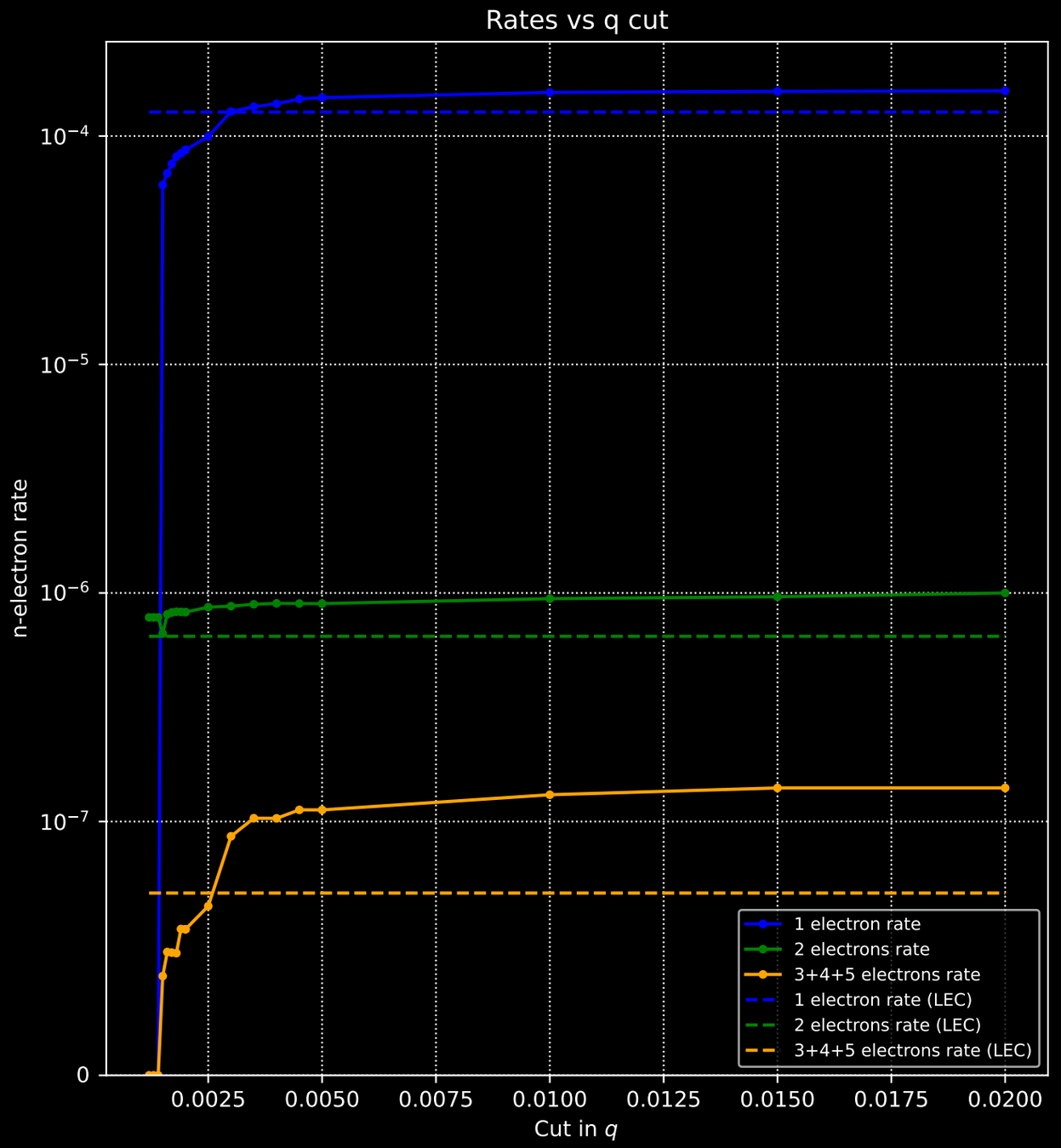


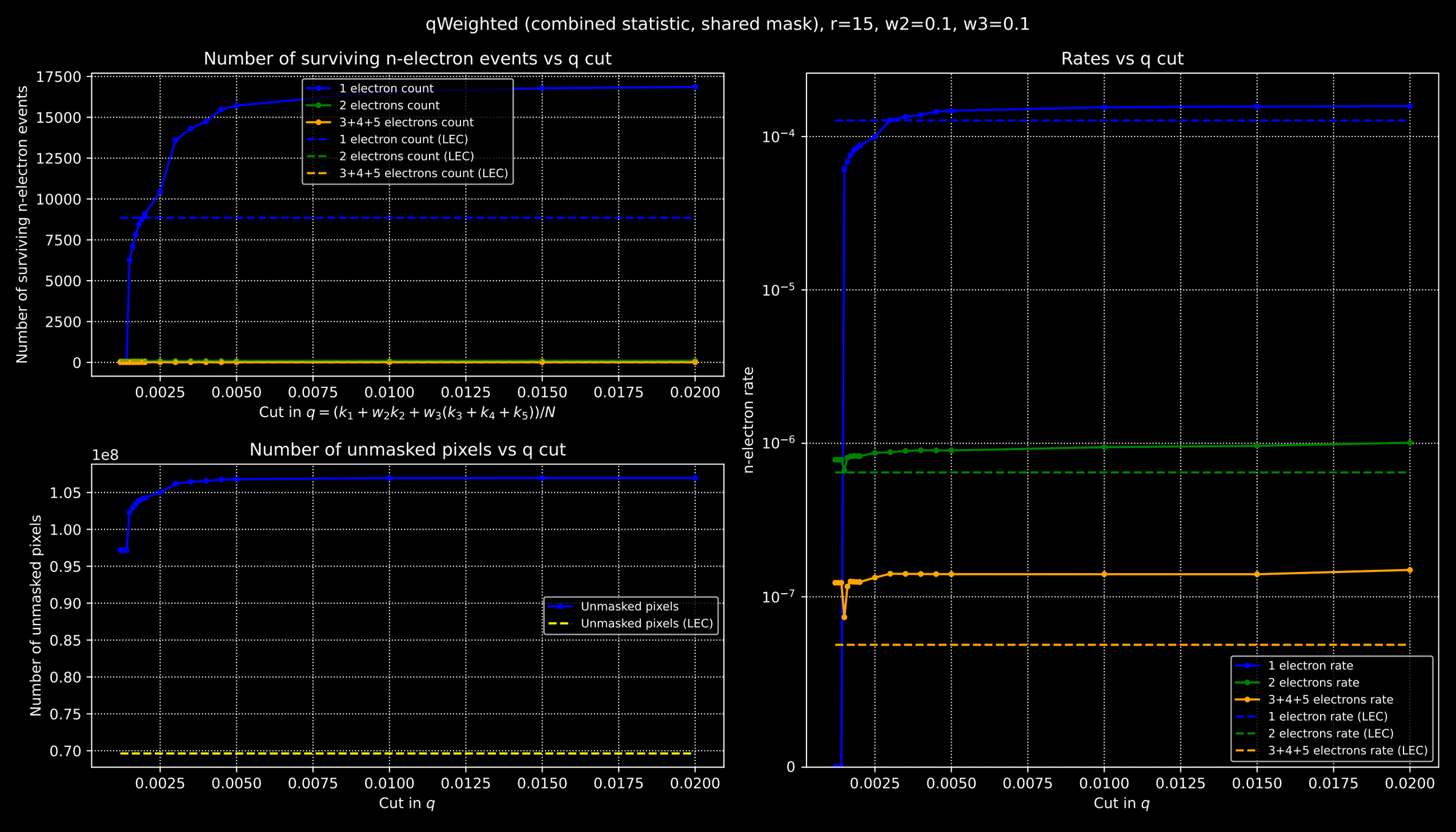


qWeighted (combined statistic, shared mask), r=15, w2=0.1, w3=1

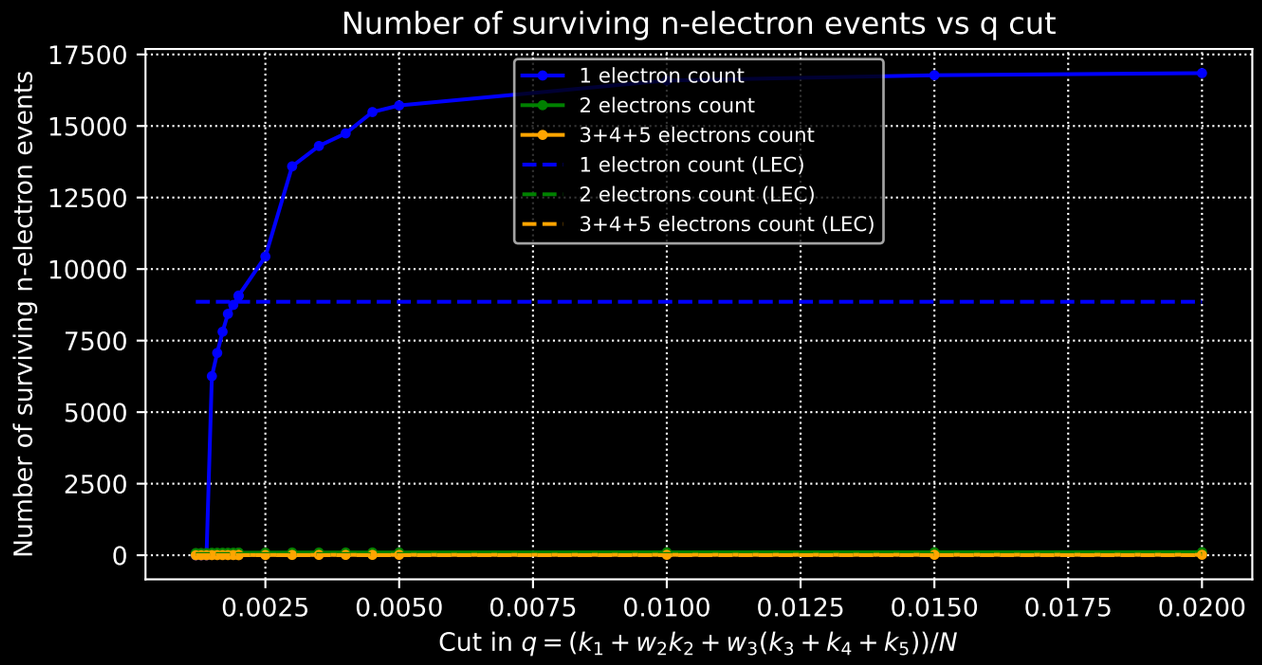


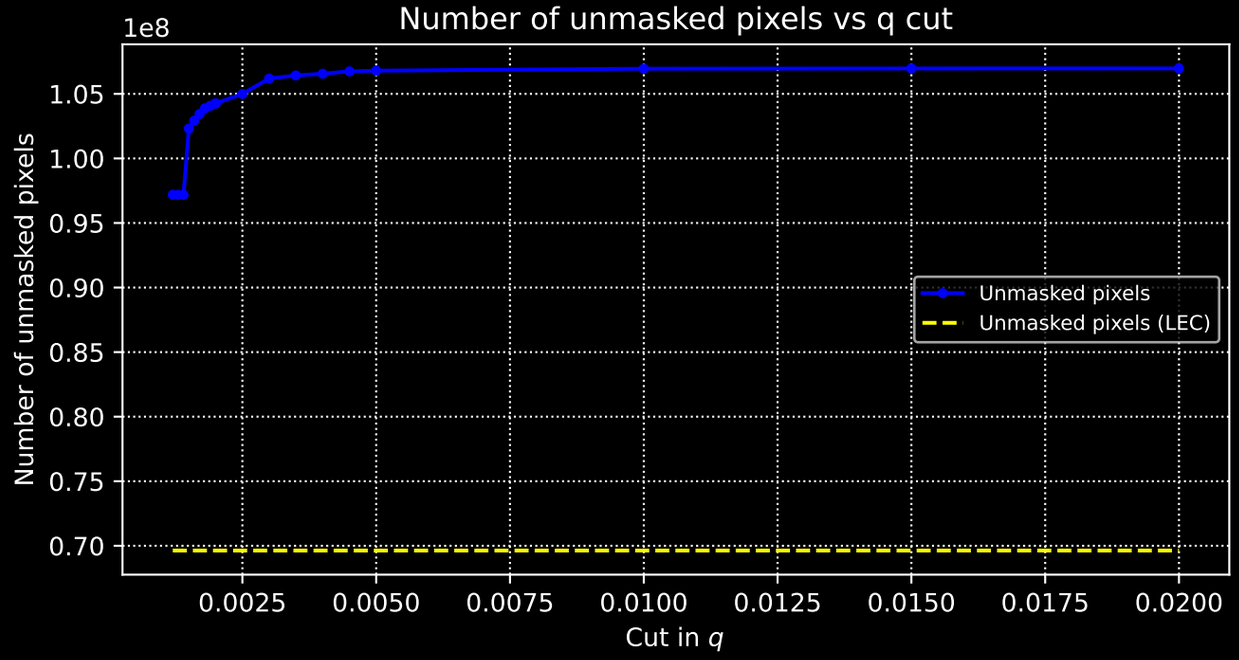


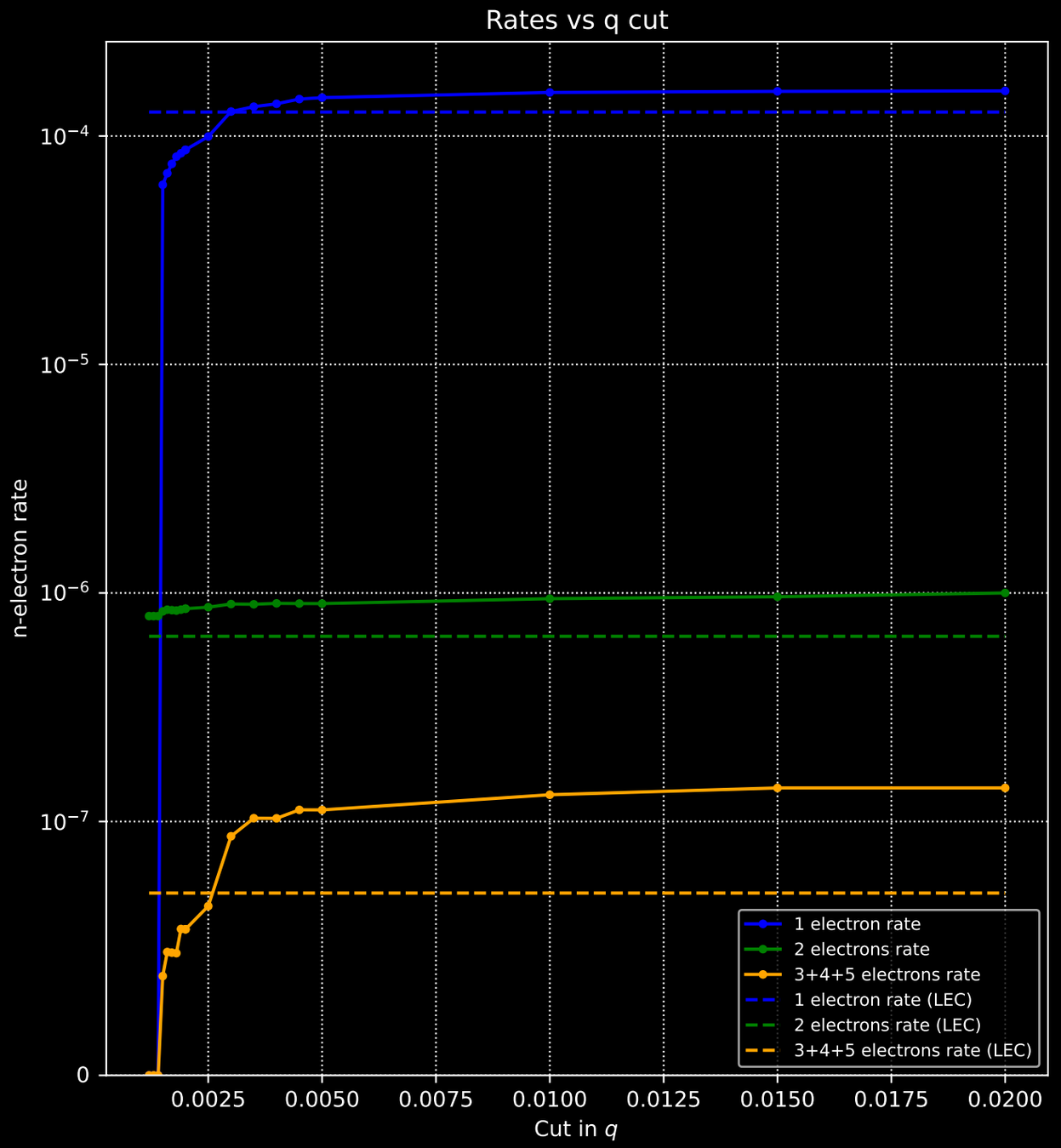



qWeighted (combined statistic, shared mask), r=15, w2=0.1, w3=0.1
Number of surviving n-electron events vs q cut
Number of surviving n-electron events
Cut in $q = (k_1 + w_2 k_2 + w_3(k_3 + k_4 + k_5))/N$
1 electron count
2 electrons count
3+4+5 electrons count
1 electron count (LEC)
2 electrons count (LEC)
3+4+5 electrons count (LEC)
Number of unmasked pixels vs q cut
1e8
Number of unmasked pixels
Cut in q
Unmasked pixels
Unmasked pixels (LEC)
Rates vs q cut
n-electron rate
Cut in q
1 electron rate
2 electrons rate
3+4+5 electrons rate
1 electron rate (LEC)
2 electrons rate (LEC)
3+4+5 electrons rate (LEC)

qWeighted (combined statistic, shared mask), r=15, w2=0.01, w3=1







qWeighted (combined statistic, shared mask), r=15, w2=0.01, w3=0.1

